



Ten questions concerning calibrating building energy simulation models

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ABSTRACT

Model calibration is routinely conducted in the building energy modeling (BEM) workflow to improve the agreement between simulation results and measured data. This process is crucial for enhancing a model's credibility before it is used for applications such as measurement and verification, retrofit analysis, fault detection and diagnosis, and building operations and control. With growing interest in using calibrated models to bridge the energy performance gap and digitalization enabling easier integration of data from multiple sources, there is an opportunity to reduce uncertainties and increase confidence in a model's predictions by calibrating with a wide range of building data inputs. However, calibration remains challenging due to the lack of established guidance and best practices. Most existing studies focus on individual case studies, and the absence of benchmark datasets and standardized evaluation methods limits progress, especially given the diversity in building types, system operation and control, and occupancy patterns. This paper presents ten key questions addressing foundational and emerging issues in BEM calibration. These questions cover the purpose and methods of calibration, challenges in handling uncertainties, necessary data requirements, evaluation metrics, and the potential impact of recent advancements in AI and machine learning. Through these questions and answers, we aim to provide a structured approach to enhance calibration practices and offer insights for current and future researchers.

1. Introduction

There are increasing concerns about the discrepancies between simulation predictions and actual measurements, commonly known as the “performance gap”, as building energy models (BEMs) are used beyond the design phase [1,2]. De Wilde [3] concluded the need for a coordinated approach to better understand and bridge the performance gap, advocating for comprehensive evidence collection, improved data inputs, and more rigorous model calibration practices.

Given the persistent and complex nature of the performance gap, it is worth reflecting on the well-known quote by George Box, “All models are wrong, but some are useful” [4], which underscores the importance of acknowledging modeling limitations while striving for utility in actual applications. Model calibration, discussed as one of the ten challenges of building energy modeling and simulation [5], serves as a critical step in addressing discrepancies by systematically adjusting uncertain parameters within a BEM to better align simulation predictions

with actual measurements [6]. This calibration process is distinct from model verification, which ensures that the implementation accurately reflects the developer's conceptual description, and validation, which assesses the model's accuracy as a real-world representation relative to intended uses. Despite its importance, BEM calibration remains under-standardized, often impeded by limited guidance, insufficient or non-informative data, and ambiguity regarding appropriate methods and metrics.

This ten questions paper addresses critical foundational and emerging issues in building energy model (BEM) calibration through ten interrelated questions (Fig. 1). It begins with questions that establish the motivation for calibration and how its performance is evaluated (Q1–Q2). The next set of questions examines the requirements for weather data and thermal zoning, calibration approaches, and the available open-source tools and datasets that support the process (Q3–Q5). This is followed by a discussion of modeling components and

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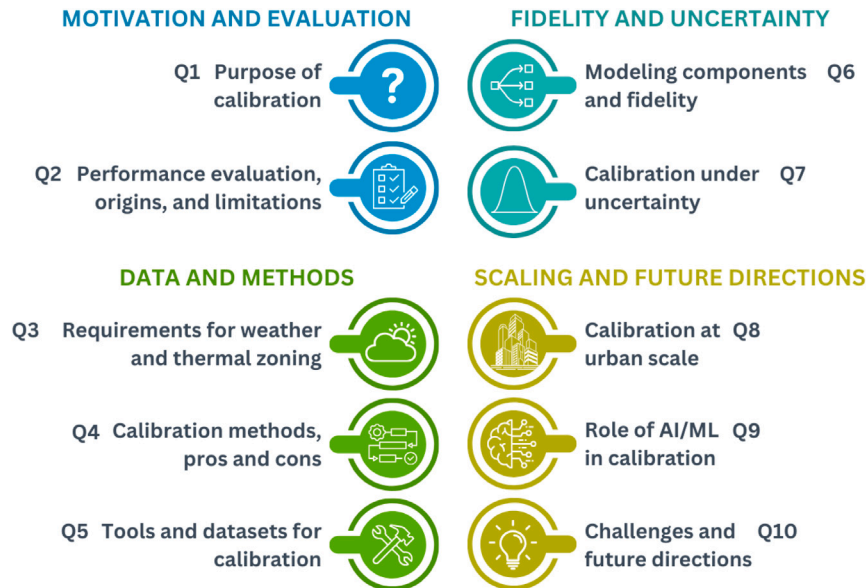


Fig. 1. Overview of ten key questions and answers on calibrating building energy simulation models, grouped into four categories: motivation, data and methods, fidelity and uncertainty, and future directions.

fidelity (Q6), and the sources of uncertainty and their treatment in calibration (Q7). Question 8 compares calibration practices between individual buildings and urban-scale models. Finally, the paper explores the role of emerging methods and outlines key challenges and future directions for research and practice (Q9–Q10). By critically examining these topics, the paper aims to clarify current practices, consolidate established and emerging methods, and propose future directions for both researchers and practitioners, ultimately contributing to the credibility and advancement of BEM calibration.

2. Ten questions (and answers) concerning calibrating building energy simulation models

2.1. Question 1: Why should we calibrate building energy simulation models?

BEMs have been widely employed to predict energy consumption and demand profiles, informing decisions on energy efficiency, control strategies, and operational improvements throughout the life cycle of buildings. In recent years, BEMs have also been embedded as an essential component in digital-twin frameworks, where they are supposed to accurately capture the bidirectional information flow and decisions between the physical world and the digital world, enabling many applications to improve building operations and performance [7]. However, discrepancies frequently arise between the simulated and measured energy use, a phenomenon often referred to as the “performance gap”. The process and complexity to create BEMs involve unavoidable simplification, approximation, and inherent uncertainties (details in Q4). For example, BEMs built at the design stage cannot capture post-occupancy operational factors such as occupant behavior [8], equipment performance drift or faulty operations [9], actual local weather conditions [10,11], and operational variability [12], resulting in significant performance gaps [13]. Through the lifetime of buildings, operational factors such as occupant schedules, thermostat settings, plug loads, and infiltration rates can vary over time as building conditions change, leading to performance gaps [14,15]. These performance gaps would reduce the model’s credibility and users’ confidence in making decisions based on the simulation results. Model calibration,

identified as one of the top ten challenges of BEM [5], aims to reconcile these discrepancies by adjusting uncertain model parameters so that the simulation closely reflects actual energy consumption, thus providing a more convincing basis for performance evaluations and investment decision making.

So, what are the major use cases of calibrated BEMs? This question introduces the primary applications of BEM calibration.

Building retrofit analysis. A prominent application of calibrated models is evaluating energy retrofit measures. Before-and-after comparisons depend on a baseline model that accurately represents pre-retrofit operations [13,16,17]. By calibrating the baseline model to real measurements, analysts can better estimate how a retrofit measure (e.g., a proposed insulation upgrade or HVAC replacement) will affect energy demand or consumption. After the retrofit, measured savings can be compared to the calibrated baseline’s predictions, helping to confirm whether savings targets could be achieved. By ensuring the model’s reliability, investors and stakeholders can make more confident and accurate decisions regarding retrofit investments.

Measurement & Verification (M&V). M&V quantifies and confirms the actual energy savings resulting from the implemented energy retrofits by comparing post-retrofit performance to a reliable baseline. Among the several methods available for establishing M&V baselines, calibrated BEM are useful when baseline energy data are unavailable or unreliable, when isolating the impact of individual energy efficiency measures (EEMs) is costly, or when complex interactions among EEMs make traditional isolation techniques impractical [18]. As errors in the baseline model can obscure or inflate perceived savings, model calibration plays a pivotal role in M&V by establishing trustworthy baselines [19–21]. Calibrated models also align with formal M&V protocols such as the ASHRAE Guideline 14 [22] and the International Performance Measurement and Verification Protocol (IPMVP) [18], which suggest thresholds for model accuracy that are intended to guide the development of reliable baselines.

Building control optimization. While it is challenging to test and optimize control strategies in occupied buildings, calibrated BEMs can capture the dynamics in building systems and thus serve as a high-quality testbed without interfering with buildings’ normal operation.

Model predictive control (MPC) is a common use case, which adopts calibrated models to optimize system operations by predicting near-future conditions (weather changes, occupancy levels, etc.) [23–25]. By factoring in energy prices or user comfort preferences, an MPC system uses the calibrated model to test thousands of potential control sequences quickly, identifying the best approach to minimize costs while not sacrificing comfort. Calibrated models have also been used in reinforcement learning controls to optimize building performance [26, 27].

Commissioning and fault detection and diagnosis. Commissioning seeks to ensure that building systems function as designed—particularly critical in complex HVAC installations. A calibrated BEM acts as a digital twin or “virtual reference” for normal building performance [28,29]. Deviations of measured consumption from model predictions may indicate equipment faults or suboptimal controls [30,31]. For example, if a fan is stuck at a higher speed than intended, real consumption data may deviate substantially from the predicted baseline in certain time periods [25]. Early detection of these discrepancies can prompt maintenance checks, which prevents larger failures and unplanned outages and saves energy [32]. Calibrated models are sometimes used to simulate effects of faulty operations such as sensor errors and faulty air economizers (ref).

Energy forecasting and planning. Calibrated models, either at the individual building level, the district level, or the urban scale, support both short- and long-term energy forecasting. In the short term, a well-tuned model can accurately predict next-day or next-week consumption under varying weather conditions [32,33]. Long-term forecasting might involve sensitivity analyses of how building loads will evolve if occupancy patterns change, if new equipment is installed, or if the climate shifts. Planners use these forecasts to decide on capital upgrades, assess resilience strategies, or develop energy budgets. Without calibration, uncertainties in parameters can compound, making forecasts less reliable. A calibrated model instead provides a solid baseline from which to explore “what if” scenarios with confidence in the underlying physics and operating data.

Thermal resilience under extreme weather conditions. Existing buildings are increasingly vulnerable to extreme weather events, such as heatwaves and cold snaps, which are becoming more frequent, intense, and prolonged. These events may coincide with, or even cause, power outages that will result in an unsafe indoor environment, exposing residents to health risks [34]. In response, optimization of building performance and investment provides an opportunity to consider co-benefits of thermal resilience during energy efficiency retrofits [35]. By ensuring alignment with measurement data, model calibration significantly improves both the accuracy and reliability of building energy models. This enables robust evaluations of thermal resilience under extreme conditions — including unforeseen more extreme future scenarios — and provides reliable estimates of how mitigation measures can enhance resilience while reducing overall energy consumption, thus supporting investment decisions [36,37].

2.2. Question 2: How is calibration performance evaluated, and what are the origins and limitations of current metrics?

A systematic review of 107 BEM calibration journal papers published between 2015 and 2020 found that the Coefficient of Variation of the Root Mean Square Error (CVRMSE) and the Normalized Mean Bias Error (NMBE) were the most commonly used metrics for determining whether a model is considered “calibrated” [6]. The widespread use of CVRMSE and NMBE is unsurprising, as current industry guidelines recommend specific thresholds when using calibrated BEM simulations. Specifically, ASHRAE Guideline 14 [38] states that a model is deemed calibrated if it achieves $\text{NMBE} \leq \pm 5\%$ and $\text{CVRMSE} \leq 15\%$ with monthly data. For hourly data, the thresholds are $\text{NMBE} \leq \pm 10\%$

and $\text{CVRMSE} \leq 30\%$. While the IPMVP does not prescribe specific thresholds in its core concept document [18], it references CVRMSE and NMBE as example metrics for assessing calibration performance. Additionally, IPMVP application guides [39] include examples that align with ASHRAE Guideline 14 thresholds. Earlier versions of its concepts document from 1997 [40] and 2002 [41] also suggest CVRMSE of 10%–20% with hourly and 1%–5% with monthly data is possible.

The widespread use of CVRMSE and NMBE traces back to the Great Energy Predictor (GEP) Shootouts, held in 1993 (GEP I) and 1994 (GEP II), which evaluated statistical models such as regression and neural networks for predicting whole-building energy use [42–45] (see Fig. 2). Around the same time, early studies began adopting CVRMSE and NMBE to evaluate calibrated simulations, often referencing the GEP Shootouts as benchmarks [46,47]. In particular, Cohen and Bailey [47] found that hourly CVRMSE values between 10% and 30% were typical for large buildings, aligning with thresholds later adopted in ASHRAE Guideline 14. However, Reddy [48], as part of ASHRAE RP-1051, remarked, “No specific reason is provided as to why the above tolerance values were proposed, but they are presumably based on the practical experience of the energy modelers who performed calibrated simulations”, cautioning that these thresholds were developed primarily for regression models and lacked a rigorous statistical basis for determining whether a calibrated simulation model is “good enough”. Notably, the more recent GEP III competition shifted away from CVRMSE and NMBE, opting instead for conventional data science metrics such as Root Mean Square Error (RMSE) and Root Mean Squared Logarithmic Error (RMSLE) [49].

Fast forward to the last decade, the use of CVRMSE and NMBE has become prevalent in BEM calibration, particularly in automated calibration processes, where the number of related publications has grown significantly, driven by increased computational power and advances in algorithmic methods [6,13,48]. A key difference between earlier studies and more recent work is that although statistical metrics were used to guide calibration by minimizing residuals, earlier efforts typically combined these metrics with heuristics, short-term monitoring, and/or graphical comparative analysis of simulation predictions and measured data to ensure judicious parameter tuning [47,50–54]. In contrast, recent approaches tend to focus on developing scalable automated algorithms that target residual minimization, often relying primarily on statistical accuracy metrics to determine whether a model is “calibrated” [55–57].

CVRMSE and NMBE have also been extended to evaluate other simulation outputs, particularly indoor temperature [36,58,59]. However, caution is warranted, as a seemingly small CVRMSE or NMBE can mask practically significant deviations when assessed in absolute terms. We take the position that a more nuanced approach to calibration is necessary. Rather than relying on fixed thresholds, calibration performance should be context-dependent. Key contextual factors include the output variable of interest (e.g., energy use, zone temperature), the required time resolution (e.g., monthly vs. hourly), the spatial scope (e.g., single zone vs. whole building), and the model’s intended use (e.g., M&V, control optimization, load forecasting). These considerations affect both performance evaluation and the appropriate level of modeling detail (Table 2). Similar to the IPMVP [18], we recommend that calibration criteria be agreed upon by stakeholders based on project-specific needs.

Unfortunately, deriving alternative threshold values for specific applications is challenging, as most studies rely on existing guidelines to justify model adequacy rather than empirically evaluating whether these thresholds are appropriate. As a result, there is limited application-specific evidence to support alternative thresholds. We recommend that future work explicitly assess whether particular calibration thresholds are sufficient for different model uses.

Given these considerations, we argue that it is currently not possible to prescribe a universal set of calibration metrics and thresholds due to the non-linearity and complexity of building energy models, as well as

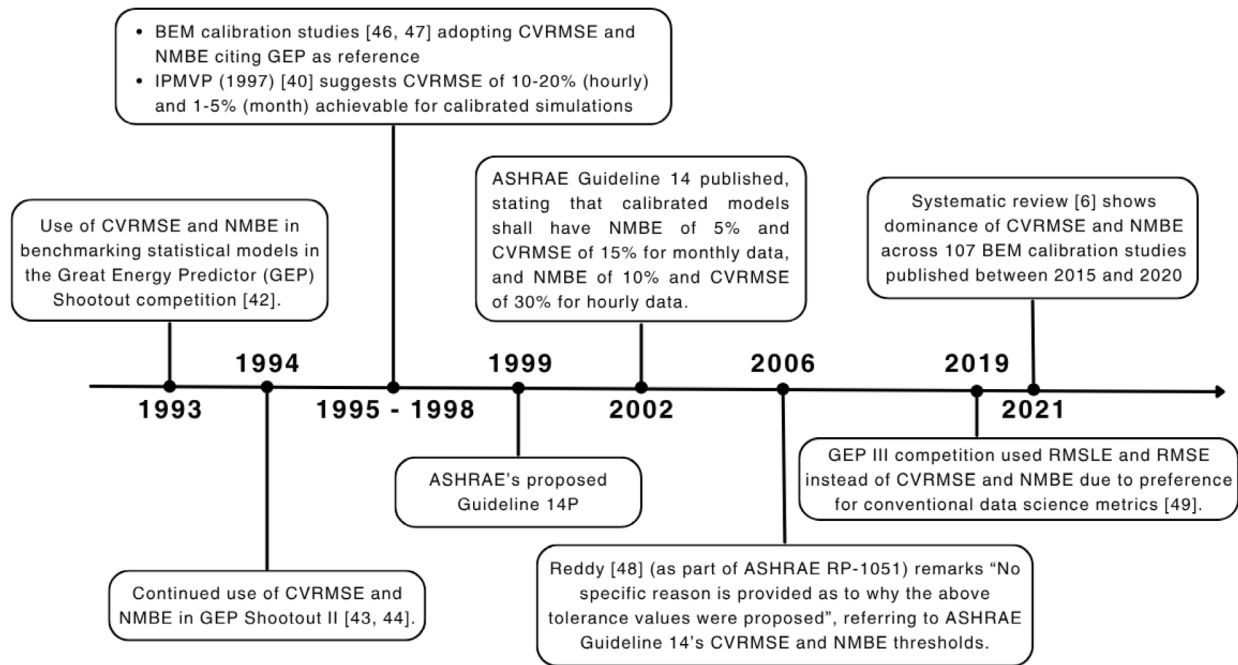


Fig. 2. Timeline highlighting key developments in the use of CVRMSE and NMBE as calibration metrics for building energy models, from their origins in regression modeling to their widespread adoption.

the diversity in building types, operational characteristics, and modeling objectives. Consequently, rather than defining universal criteria for determining when a model is “calibrated”, we present a structured categorization of calibration metrics, highlighting their attributes and limitations while leaving the selection to modelers. To illustrate this, we provide representative examples of point distance accuracy measures, time-series similarity metrics, and probability-based evaluation metrics (Table A.3). These examples are illustrative, not exhaustive. For more comprehensive coverage, we refer readers to [60–62].

2.3. Question 3: What weather data and zoning configurations are appropriate for BEM calibration?

Among the many components that structure a BEM, weather data and thermal zoning are often the first considerations when developing a model [63]. Weather data defines the external boundary conditions and directly influences the energy balance at external surfaces, while the thermal zone, as the basic geometrical and thermal building block, directly affects input data requirements, simulation accuracy, and troubleshooting complexity. Because these inputs frame how other subsystems and loads are represented, simplifications in weather data and zoning warrant careful attention in the modeling and calibration process.

Weather data. Typical Meteorological Year (TMY) [64] and the Actual Meteorological Year (AMY) [65] are the two common types of weather files used for whole building energy simulations.

There are some debates on whether TMY weather data can be used for BEM calibration. For example, when using a calibrated simulation approach for measurement and verification, ASHRAE Guideline 14 [66] emphasizes that actual weather data during the reporting period must be collected to match the metered or sub-metered data at monthly or hourly temporal resolution. However, IPMVP [41] permits the use of TMY weather data when calibrating against monthly utility data.

Debates over the appropriate weather data for building energy simulations date back to the late 1990s, with early comparative studies by Crawley [67] for commercial buildings and Huang [68] for residential buildings. Both studies found that simulations using TMY2 and WYEC2 (Weather Year for Energy Calculations) datasets produced annual energy use estimates within 5% of the 30-year AMY averages. Since then, a growing body of research has continued to compare simulation results based on TMY and AMY data [69–77]. Key findings from these studies are summarized below.

- **Building level:** At the whole-building level, annual energy use differences between TMY and the multi-year average of AMY results generally range from under 5% to around 10% [67–69,72]. However, deviations from specific AMY years can be substantially larger. Some studies also report differences of up to 44% when comparing on-site weather data with that from nearby stations, highlight the impact of local microclimates [71,75,78,79].
- **System/Sub-system level:** At the system or sub-system level, weather-driven discrepancies are often more pronounced, particularly for HVAC loads. Hong et al. [72] reported a 37% difference in annual HVAC energy use when using TMY versus AMY weather data. Cui et al. [73] found that TMY overestimated 55-year AMY averages by up to 7% for total HVAC energy use, and by 18% and 13% for cooling and heating, respectively. Because heating and cooling errors may offset one another, whole-building totals can understate sub-system discrepancies [71].
- **Temporal resolution:** While most studies compared annual energy use differences, those investigating time-series and point-in-time deviations (e.g., peak loads) reported substantial differences up to 69% [67,72,76,80], indicating that aggregating to annual values may obscure short-term discrepancies introduced by TMY data.

Thermal zoning. The level of detail in thermal zoning defines the spatial granularity of building energy models, and may vary depending on the simulation’s intended use. ASHRAE Standard 209 [81] recommends

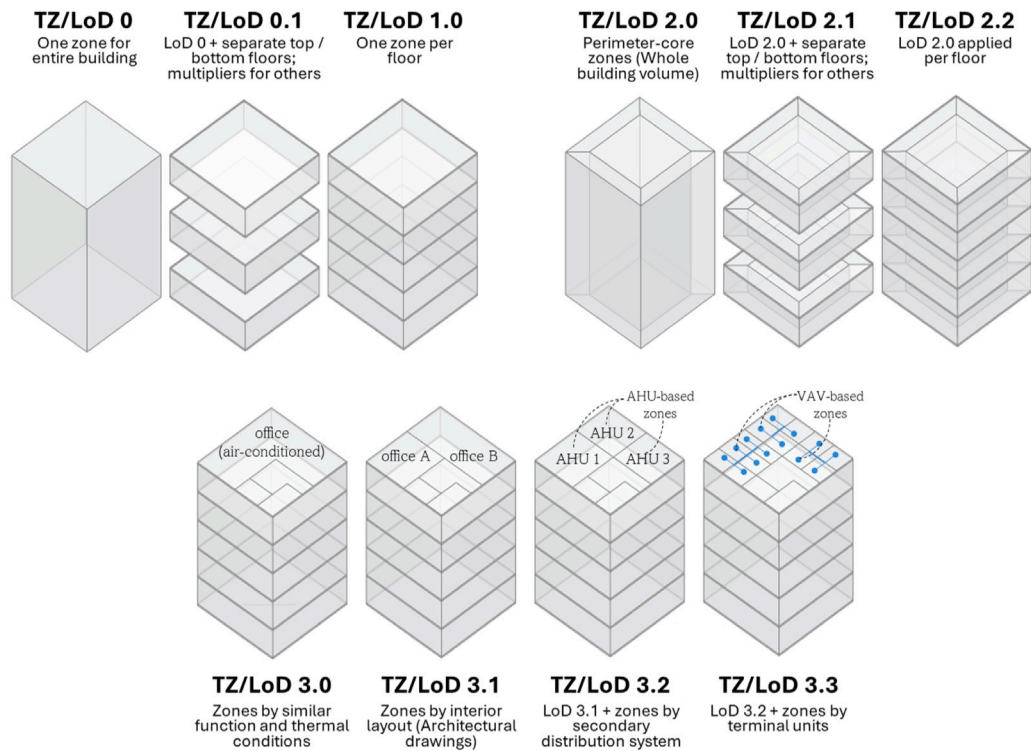


Fig. 3. Levels of detail for thermal zoning (TZ/LoD), compiled and adapted from existing literature.

a perimeter-core zoning approach [82] for simple box models during early design. More detailed models may use one zone per room when interior layouts are available [83]. For buildings with repetitive floors, zone multipliers are commonly applied [84], and ASHRAE Standard 90.1 [82] recommends modeling the top and bottom floors separately from intermediate ones.

When HVAC systems are defined, thermal zoning configurations are often aligned with the level of HVAC modeling. Zones served by the same system and exhibiting similar thermal conditions may be aggregated into a single thermal block, or subdivided to match areas served by individual air-handling units or terminal devices. Fig. 3 summarizes zoning strategies and the corresponding levels of detail for thermal zoning (TZ/LoD) compiled from the literature. The TZ/LoD categorization used in this paper is adapted from the multidimensional specification proposed by Dilsiz et al. [85], which extends the original LoD concept from the BIM context [86] to standardize model resolution descriptions and enable comparison across BEM studies.

Several studies have assessed the impact of thermal zoning on model accuracy, with TZ/LoD 3.1 [83,87,88], TZ/LoD 2.2 [89,90], and TZ/LoD 1.0 [91,92] being the most frequently compared. Key findings from these studies are summarized below.

- **Energy end-use:** Thermal zoning simplifications tend to result in discrepancies in heating and cooling energy use, as under-zoning can neglect localized heat gains and thermal variation, leading to inaccuracies in load predictions, with reported impacts ranging from an average of 3% for annual cooling to 26% for annual heating [89,91,93].
- **Zone temperature:** Thermal zoning can significantly impact zone temperature predictions. Between TZ/LoD 2.2 and TZ/LoD 3.1, discrepancies of up to 1.6 °C have been reported [94]. Finer zoning has been shown to improve control performance and

reduce unmet setpoint hours from 30% to below 5% [87]. Zoning based on terminal units has also been shown to reduce flow rate errors by 28% [95]. These findings support the use of TZ/LoD 3.1 for accurate zone temperature prediction and TZ/LoD 3.3 for high-resolution HVAC simulations.

- **Temporal resolution:** Thermal zoning has a more pronounced impact on time-series model accuracy than on annual aggregates. Across three studies comparing TZ/LoD 2.2 and TZ/LoD 3.1, Kong et al. [88] reported an hourly CVRMSE of up to 81% in total energy, Bres et al. [87] observed daily CVRMSE values of up to 31% in heating energy, and Georgescu et al. [83] concluded annual differences of up to 8% in HVAC energy use.

Despite these observed discrepancies, the selection of weather data and thermal zoning should match the simulation's intended purpose. A potential framework is the principle of non-inferiority, which originates from clinical research where it is used to assess whether a new treatment performs no worse than a standard option by more than a predefined margin [96]. Applied to BEM calibration, this suggests that simplifications such as coarser weather data or zoning may be justified if their performance remains within an acceptable margin relative to a more detailed alternative.

2.4. Question 4: What are the existing calibration methods and their pros and cons?

Existing methods to systematically adjust calibration parameters can be categorized into evidence-based, optimization-based, and calibration under uncertainty. The three categories are distinguished from each other in data sources and usage, as well as the calibration process. Table A.1 summarizes the distinct characteristics of the three categories, highlighting the differences in data/knowledge requirements and the key advantages and limitations.

Evidence-based calibration. This is an expert-driven process that uses building information such as as-built drawings, system specifications, audit findings, and short-term measurements to inform parameter adjustments. It is typically a manual, forward process guided by documents, standards, and measurements. For example, building metadata, such as refined usage patterns, can be used to update assumptions from reference models and regional codes [97], while as-built drawings and system specifications can inform envelope properties and HVAC system capacities [98]. Site visits often provide additional data through audits, interviews, and in-situ measurements [99], and operational schedules may be derived from short-term monitoring [100].

On-site measurements of observable model inputs can play a critical role in improving parameter identifiability. Table A.4 summarizes commonly used methods for measuring observable model inputs, as outlined in international standards. These include the determination of thermal resistance and transmittance (BS ISO 9869-1 [101]); measurement of infiltration rates using tracer gas dilution (BS EN ISO 12569 [102]) or fan pressurization techniques (BS EN ISO 9972 [103]); and the monitoring of occupancy using various methods recommended in IEA EBC Annex66 [104]. For non-observable parameters, the process may be treated as an inverse problem, inferring values from measured outputs. For instance, zonal infiltration rates and thermal mass have been estimated from indoor air temperature measurements [105], and other studies have tuned parameters through trial and error to match hourly temperature profiles [106].

Some steps can be automated using methods such as pattern-based bias analysis [54]. Although grounded in static information or snippets of dynamic data, the incorporated expert knowledge offers relatively high credibility and physical representativeness of the calibrated parameters [107]. However, the selection of calibration parameters depends on data availability, and model performance is constrained by data quality. Without sufficient input data, the calibrated model may fail to meet accuracy requirements [108]. Therefore, evidence-based calibration is particularly valuable when expert input is available and building information is readily accessible. In addition to serving as a stand-alone method, it is commonly used to refine the model before applying automated approaches [109]. Even when not directly involved in an automated approach, static building information can support calibration by providing a more reliable base model.

Optimization-based calibration. This is an automated approach that identifies the optimal set of parameter values that minimize the objective function, which is usually defined as the discrepancy between model output(s) and the corresponding measured data, given a period of observed inputs. Particle swarm optimization [110–112] and genetic algorithm [56,113–115] are the two most widely used algorithms, respectively inspired by natural evolution and social behavior. These population-based heuristic optimization methods are gradient-free and, therefore, suitable for building energy models that make the objective function non-derivative. Other more data-efficient algorithms, such as Bayesian optimization [116], could also be adopted to reduce computational cost. However, as the number of parameters to calibrate increases, the required number of simulations could still drastically increase, which is especially problematic for large and complex BEMs. Hence, it is common to combine surrogate models (also referred to as meta models) with optimization-based calibration, which learns the input–output relationships from an initial batch of simulation results and then is used to replace the simulations for further iterations [30, 57,117,118].

As an inverse problem, the primary data requirements of optimization-based calibration are the observed model inputs and outputs, including weather conditions, electricity [95,119], heating and cooling loads [120,121], room temperature [59,122], etc. The optimization can simultaneously calibrate against multiple outputs [123], using measured data with higher resolution or spanning a longer period. Meanwhile, although the algorithms themselves are automated,

experts are still critical to properly create the base model, set up the optimization problem, and define the admissible parameter range. For example, it is inadvisable to calibrate many parameters in one optimization task. Despite the potential low predictive error, the physical meaning of parameters could be compromised in such an ill-posed problem. The calibrated parameters should be carefully selected, by experts and/or through sensitivity analysis (SA). As a crucial supporting tool, more information about SA is introduced in Question 5. To summarize, compared with evidence-based approaches, optimization-based calibration could make use of a larger operational dataset and achieve a lower error. Yet, expert knowledge plays an important role in defining the optimization task so that the parameters are identifiable and reliable. In cases where many parameters are uncertain and correlated data is available, a multi-stage approach could be adopted to categorize the parameters and calibrate them in different batches with sequential optimization tasks [123–125].

Calibration under uncertainty. The calibration of building energy models is challenged by multiple sources of uncertainty, including unknown or variable model parameters, structural discrepancy between the model and reality, and measurement noises or errors [6,126]. Bayesian methods are commonly used in this context, combining prior knowledge with measured data to estimate posterior distributions of uncertain parameters. While similar to optimization-based approaches in data requirements, Bayesian methods introduces two distinct differences. First, sampling techniques such as Markov Chain Monte Carlo are typically used to approximate posterior distributions [127,128], but the computational complexity of full Bayesian calibration is $O(n^3)$, making it intractable for large datasets [129]. Therefore, surrogate models such as Gaussian processes and linear regression have been employed to approximate the model response at lower computational cost [130–133]. More efficient sampling strategies have also been proposed to reduce computational costs [134,135]. Second, Bayesian calibration requires significant expert involvement, including the specification of prior distributions [130], design of likelihood and error models [131], and interpretation of trace diagnostics for correlated outputs [136]. For a detailed discussion of calibration under uncertainty, see Question 7.

2.5. Question 5: What open-source tools and datasets are available to support model calibration?

Open-source tools and publicly accessible datasets for BEM calibration are becoming increasingly available. These resources help to facilitate a more efficient, streamlined, and reproducible calibration process. This section provides an organized overview of current tools, datasets, and relevant standards, categorized according to their roles in the calibration workflow.

Open-source tools. These tools are categorized according to their functions. Considering that current BEM calibration workflows are primarily implemented in Python or R, representative packages in both languages are summarized in Table 1.

Sensitivity analysis, to identify the key variables during BEM calibration, can be divided into local and global methods. Global method examines how uncertainties in a model's output can be attributed to the uncertain inputs across the entire input space [137]. As for local sensitivity analysis, it assesses sensitivity by measuring variations in model outputs around a baseline case, with inputs perturbed within predefined ranges [138]. Details of the tools can be found in the table.

To rapidly obtain simulation results with low computational cost, surrogate models have been introduced. Existing studies have introduced various open-source tools that facilitate surrogate model development. In Python, Sklearn [139] is a widely-used package, including multiple machine learning (ML) models. Correspondingly, Caret [140] and Nnet [141] in R language can help build ML models. Also, as mentioned in Q4, optimization-based calibration requires

Table 1
Open-source tools for BEM calibration.

Description	Tool type	Language	Package	Ref.
To identify key variables influencing the target building performance	Sensitivity analysis	Python	SALib: Sobol, Morris, FAST	[142]
			pyDOE: LHS	[143]
			SciPy.stats (local sensitivity)	[144]
		R	sensitivity: Sobol, Morris, FAST lhs: LHS FME: local sensitivity	[145] [146] [147]
To create substitute models for high-fidelity building energy models, reducing computational cost without compromising much accuracy	Surrogate model	Python	scikit-learn: RF, XGBoost, SVM	[139]
			Keras: Deep learning structures	[148]
			GPY: Gaussian Process Regression	[149]
		R	caret: ML regression models nnet: Neural networks	[140] [141]
To automate the use of mathematical methods for minimizing the errors between simulation and reality	Optimization algorithms	Python	SciPy.optimize	[144]
			DEAP: Evolutionary algorithms	[150]
			PySwarm: PSO algorithm	[151]
		R	BB: GA, PSO GA, DEoptim, PSO (individual packages)	[152] –
To derive posterior distributions of uncertain parameters	Bayesian calibration framework	Python	PyStan (high resolution), PyMC (simpler)	[153,154]
		R	rstan, abc, calibrator	[155–157]

Table 2
Three dimensions to define BEM fidelity and relevant key modeling components for model calibration.

Dimensions	System boundaries	Spatial resolution	Temporal resolution
Model fidelity levels	- Demand side – Spaces - Thermal loads - Supply side – HVAC - Primary systems - Secondary systems	- Sub-building - Building - Building stock (district/urban/regional/national)	- Static snapshot (CFD) - Aggregated - Annual - Seasonal - Monthly - Time series - Daily - Hourly - Sub-hourly
Relevant key modeling components	- Demand side – Spaces - Internal heat gains (lighting, plug loads, occupants) - Envelope properties - Geometry - Supply side – HVAC - Detailed HVAC systems (component performance, operation and control strategies)	- Sub-building & Building - Internal heat gains - Envelope properties - Occupant behaviors - Mechanical equipment and controls - Non-uniform air distribution (Sub-building only) - Detailed zoning (Building only) - Centralized HVAC systems (Building only) - Building stock - Distribution of building characteristics	- Static snapshot - Non-uniform air distribution - Seasonal - Cooling/heating setpoints - Cooling COP/heating efficiency - Timeseries - Equipment operation schedules - Occupancy schedules - Occupant behaviors

specific algorithms. Several commonly adopted open-source packages are organized in Table 1.

For calibration under uncertainty, it usually involves specific Bayesian Calibration Frameworks. In Python, *pystan* and *pymc* are two commonly applied packages. The former one fits for complex and high-resolution models, while the latter one is suitable for less complex models. In R language, *rstan* is the most commonly used package [130]. Other packages including *abc* and *calibrator* are also widely adopted.

Open-access datasets. Public datasets are categorized in this study based on their functions mentioned in Q3, including weather data, basic building models, internal heat gains, and energy-use datasets for benchmarking or reproducibility. Detailed information has been organized in Table A.2.

For open-source weather data, ERA5, a widely used atmospheric reanalysis database [158], has been adopted in recent studies as the Actual Meteorological Year (AMY) for the BPS outdoor boundary inputs. It provides global coverage at a 31 km resolution, offering hourly data from 1970 to the present. Similarly, MERRA-2 is another reanalysis open dataset produced by NASA, providing hourly weather data

from 1979 till now [159]. For applications requiring higher temporal resolution or real-time weather forecasts, APIs such as MeteoBlue, OpenWeatherMap, and Open-Meteo are provided.

Public building model datasets are often applied in urban-scale energy modeling calibration. Many countries and institutions have released prototype building models. In the U.S., the DOE provides 16 public and 2 residential building models that reflect existing building characteristics and built years [160], facilitating further analysis. In China, An et al. [161] released a prototype building database covering 4 residential and 11 commercial building types, and considering 270 cities and multiple years of design codes, 9225 models are provided in total. Europe's TABULA project offers exemplary residential building models from 20 countries [162].

Beyond building typologies, several datasets on occupancy and occupant behavior have been made public to support the development of internal heat gain models for BEM calibration. Ideally, BEM calibration should be based on real, measured internal heat gain data specific to the building of interest. Yet, when such data are unavailable, these public datasets offer valuable reference points. Dong et al. [163] led the development of the ASHRAE Global Occupant Behavior Database, compiling 34 field-measured datasets from 15 countries and 39 institutions. In

addition, many countries have released time-use surveys that document daily occupant behavior [164]. The ROBOD (Room-level Occupancy and Building Operation Dataset) includes 181 days of data collected every 5 min in five office rooms [165].

Regarding energy-use datasets for benchmarking or reproducibility, the data can be broadly categorized into building energy-use data and cross-sector datasets. The former includes the ASHRAE Great Energy Predictor III dataset [49], which provides hourly energy usage data, including chilled water, electricity, hot water, and steam from over 1000 buildings across three years. The Building Performance Database (BPD) is a public data platform, enabling exploration of relationships between energy use intensity (EUI), carbon emissions, and building characteristics [166]. The Building Benchmark Datasets project curated 13 datasets with high-resolution and high-quality data, as well as metadata for buildings, systems, and monitored data points [167]. Multidisciplinary datasets usually span sectors such as transportation, industry, and agriculture, for example, the Pecan Street Dataset [168] and the Open Energy Data Initiative [169]. These cross-sector datasets include some detailed end-use breakdowns of energy consumption at the device level. For instance, the OEDI includes ground-source heat pump data, which can be useful for researchers aiming to calibrate models involving geothermal HVAC systems.

Standards and guidelines. To facilitate and ensure reproducible BEM calibrations, as mentioned in Q3, standards and guidelines have been published.

RESNET Standard 1201: Standard Method of Test for the Evaluation of Building Energy Analysis Model Calibration Methods [170] focuses on how well these models predict energy performance. The standard provides a framework for evaluating the accuracy and reliability of building energy simulation models. It focuses on the calibration process, which involves adjusting model parameters to better reflect real-world energy performance. ASHRAE Guideline 14 [22] and IP-MVP [18] are two formal M&V protocols including model accuracy thresholds, to guide the development of relatively reliable calibrated BEM as introduced in Q1.

In summary, existing studies and organizations have released a variety of open-source datasets, tools, and standards to support BEM calibration. This answer aims to organize and summarize part of these resources for easier adoption by future researchers.

2.6. Question 6: What are the main modeling components and fidelity challenges in representing reality of buildings and their systems?

A BEM is usually developed for specific applications that require particular level of details in modeling [171]. Therefore, BEM calibration should adopt the principle of fit-for-purpose [172]. Clearly defining the target and boundary of BEM calibration is critical for identifying key modeling components and their appropriate level of details. Three dimensions should be considered when defining the calibration target and boundary: the system dimension, the spatial dimension, and the temporal dimension. Table 2 summarizes the model fidelity under each dimension and their relevant key modeling components.

The system dimension defines the boundary by considering both the demand and supply sides. The demand side refers to meeting space thermal and ventilation loads, which include internal heat gains (lighting, plug load, and occupants), envelope-related heat gains (solar heat gains through windows, conduction through windows and opaque walls/roofs/floors, and air infiltration), and loads from conditioning outdoor ventilation air. The supply side comprises the HVAC system, which can be further divided into primary (plant) and secondary (distribution) systems. Space loads are usually calculated using heat balance methods, such as state-space matrices or conduction transfer functions, to account for current and past heat fluxes within the thermal zone. In contrast, HVAC systems are often modeled using more empirical approaches. While some secondary components (e.g., heat

exchangers) are based on heat transfer principles, many secondary and primary components such as chillers, boilers, fans, and pumps are often represented by regression-based models derived from manufacturers' performance data.

Depending on the application, simulation models may apply varying levels of detail to either or both the demand and supply sides. For example, Ruiz and Bandera [173] calibrated building envelope parameters by leveraging free-floating periods, during which HVAC systems were switched off. This approach isolates the demand side, therefore requiring it to be modeled in detail. Rosato et al. [174] modeled a detailed HVAC system serving a test room in TRNSYS, with the room primarily characterized by its envelope properties. The HVAC model was calibrated with measurement data and then used for fault detection and diagnose. Martinez-Viol et al. [25] calibrated an HVAC system model that couples with a building load, whose thermal behavior was modeled via a simplified parametric representation without actually matching its geometry or construction properties.

The spatial dimension determines the scale at which a model is calibrated. For example, Chapter 19 of the ASHRAE Fundamentals Handbook [175] describes five major BEM scales: sub-building, building, district, urban, and regional or national, each requiring different simulation methods and components. At the sub-building level, calibration typically focuses on building physics, mechanical equipment, or controls within a single room or a few rooms, and may exclude the entire building's centralized HVAC system [176]. Because there are fewer parameters to adjust at this scale, it usually requires more precise room-level inputs, such as internal heat gains, envelope properties, occupant behaviors, equipment control logic. In cases of non-uniform airflow distributions, coupled CFD simulations may also be required to capture spatial variations in airflow and temperature [177,178].

In Contrast, district, urban, regional, and national scales—collectively referred to here as the “building stock scale”—face higher complexity and uncertainty, but offer greater “freedom” in modeling and parameter tuning. This is partly due to the limited data availability of building stock at large scales. For instance, detailed zoning and occupant behaviors data are often unavailable and may therefore be simplified or standardized [179]. Additionally, certain inputs (e.g., HVAC system types, occupant count and schedule) may be hard to obtain for each individual building but could be available in the form of statistical distributions at the building stock level. Therefore, rather than tuning inputs for every building, it is more reasonable and practical at this scale to adjust the statistical distribution of those inputs [180].

The temporal dimension defines the time resolution at which a model is calibrated, which can be organized into four categories: static snapshot, aggregated, seasonal, and time series. Static snapshot evaluations consider conditions at a single point in time and therefore do not require dynamic inputs, such as steady-state CFD simulations. Aggregated calibration (e.g., using annual totals) focuses primarily on overall consumption, making capturing daily operations less critical. At this level, detailed control logic and operational schedules are typically of secondary importance. Monthly calibration, while still aggregated, captures seasonal variations and thus requires input parameters that account for changes in cooling and heating energy use, such as cooling and heating setpoints, cooling COP, heating system efficiency, and outdoor air flow rate [54].

Time-series calibration (daily, hourly, or sub-hourly) aims to align dynamic simulated patterns with real-world operations. Consequently, higher-resolution simulation inputs such as detailed control logic, operational schedules, and occupancy patterns become pivotal [181]. At finer sub-hourly resolutions (e.g., minute or second), calibration often targets refined models of specific components or systems that use advanced control strategies [25,174]. Software capabilities also play a role: whole-building simulation tools like EnergyPlus offer limited support for complex control logic at sub-minute time steps, whereas

component-based platforms such as Modelica enable customizable controls and more flexible time-stepping, making them more suitable for calibration at minute or second resolutions [182].

BEM calibration aims to achieve fidelity, which refers to an accuracy level that is sufficiently representative of reality to ensure confidence in using the calibrated model for its intended purpose. For example, energy retrofit analyses typically require whole-building calibration using annual or monthly energy consumption data. In contrast, applications such as building control optimization and fault detection call for calibration at the component level (e.g., chillers or VAV fans) using sub-hourly operational data. There are two challenges in achieving the targeted fidelity: (1) data availability, quality, and uncertainty, and (2) the inherent inaccuracy and limitations of the simulation software. These challenges are discussed below.

In practice, measured data may be incomplete or missing for some periods (for instance, a day or longer due to sensor power outage). Some key modeling parameters, such as thermal mass and infiltration rate, may also be difficult or impossible to capture accurately in situ [105]. Even with established site testing methods (e.g., blower door or tracer gas tests), infiltration rates can remain highly uncertain under actual operating conditions [183,184]. Additional uncertainty arises when actual building conditions deviate from design documents, while as-built drawings and specifications are still commonly used as the default reference for model inputs. For example, equipment degrades over time, windows can experience seal or coating failures, lighting fixtures may be replaced, and so on. Further complicating matters, diverse occupant behaviors, such as comfort preferences, schedules, and usage patterns, introduce yet another layer of model uncertainty. Further details about uncertainty are discussed in Q7.

A second major challenge in achieving high-fidelity model calibration arises from the inherent inaccuracy and limitations of simulation software itself. While tools such as EnergyPlus, TRNSYS, and Modelica-based libraries provide powerful capabilities for modeling building physics, their underlying assumptions and numerical methods inevitably introduce simplifications. For instance, ASHRAE Standard 140 [185], which is widely used to benchmark and validate building energy simulation programs, has limited scope and complexity. Although it verifies core calculation methods (e.g., heat transfer through walls and windows, simple mechanical systems), it does not fully address more nuanced phenomena such as detailed moisture transport, real-world occupant behavior variations, or long-term equipment performance degradation. As a result, even software passing ASHRAE 140 tests may exhibit modeling gaps or inaccuracies when applied to the full range of real-world building conditions.

For buildings with DERs, such as on-site power generation from PV systems, EV charging stations, and energy storage systems, the required model fidelity depends on the BEM's purpose. If the aim is to optimize the DERs design and operations based on dynamic utility rates, the fidelity of these systems and components within the BEM should be determined considering the calibration requirements and the availability of measured performance data.

As a summary, to determine the key components in a building energy model, three dimensions (system, spatial, and temporal) should be considered comprehensively to ensure the level of detail aligns with the model's intended use purposes. Clearly defining these boundaries helps identify which elements require the most precision. Major fidelity challenges arise from data limitations and uncertainty as well as inherent constraints in simulation tools, which may not fully capture real-world building operations.

2.7. Question 7: What are the main sources of uncertainties, and how is calibration performed under uncertainty?

BEM inherently involves uncertainty due to its broad applicability across diverse building designs and operational scenarios. As a result, BEM simulation programs often consists of large codebases

designed to capture complex interactions between multiple systems and subsystems, each subject to varying levels of abstraction and approximation, leading to representations that are invariably imperfect. Consequently, uncertainty quantification plays an important role in assessing uncertainties associated with both model inputs and predictions, providing better understanding that supports more reliable and informed decision-making [186].

Literature on the interpretation of uncertainty has often differentiated between aleatory (irreducible) and epistemic (reducible) uncertainty. Aleatory uncertainty refers to uncertainties caused by inherent randomness that cannot be reduced by collecting additional information, while epistemic uncertainty arises from a lack of knowledge and can be reduced as more information becomes available. However, it has been argued that, although this distinction has theoretical merit, it has limited practical utility for decision-making in real-world applications, as most uncertainties involve a combination of both aleatory and epistemic components [187].

Occupancy modeling illustrates how aleatory and epistemic uncertainty can coexist. Some uncertainties can be reduced through survey or sensor data, but occupant behavior involves inherent randomness that cannot be fully captured with current measurement and modeling approaches [188,189]. Interestingly, this also implies that the degree of aleatory uncertainty is influenced by the functional form of the model and that aleatory uncertainty will always be present unless exact predictive models of building energy performance exist. Moreover, what is treated as aleatory may shift over time as new data becomes available, making uncertainty partly time-dependent [190].

Nonetheless, from a model calibration perspective, distinguishing between aleatory and epistemic uncertainty remains beneficial, not as a philosophical debate in defining aleatory uncertainty, but as a practical means of identifying which uncertainties can be reduced and which are less prone to reduction, at least in the near term [191]. Such categorization can help prioritize resource allocation for sensing infrastructure and guide the development of fit-for-purpose building energy models. An alternative perspective in modeling and simulation classifies uncertainty based on its sources [192], an approach that has also been adopted in BEM [6]:

- **Parameter uncertainty:** Uncertainty associated with parameter values that are not known with certainty.
- **Model form uncertainty:** Also referred to as model discrepancy or model inadequacy, this arises from assumptions, abstractions, and approximations in the model's functional form when representing real-world physical processes.
- **Observation uncertainty:** Uncertainty in measured or observed data.

Uncertainty quantification can broadly be categorized into forward and inverse methods. Forward methods quantify uncertainty in a model's outputs by propagating uncertainties from input parameters. The most widely used forward approach in BEM is Monte Carlo simulation, which involves defining probabilistic distributions for uncertain input parameters, sampling from these distributions using random or Latin hypercube sampling, running simulations, and analyzing the resulting probabilistic outputs [193]. Non-sampling-based uncertainty propagation methods, such as generalized polynomial chaos and polynomial chaos expansion, have also been proposed to approximate uncertainty propagation in complex systems like BEM [193,194]. While forward approaches (particularly sampling-based methods) have been used to screen models that meet a specific error criterion [32,195], they do not fall within the calibration under uncertainty paradigm. In these cases, while uncertainties are considered, the approach does not fully characterize how uncertainty propagates throughout the model.

Inverse approaches such as Bayesian methods fully characterize parameter uncertainty by deriving a probability distribution over possible parameter values based on the likelihood of matching model

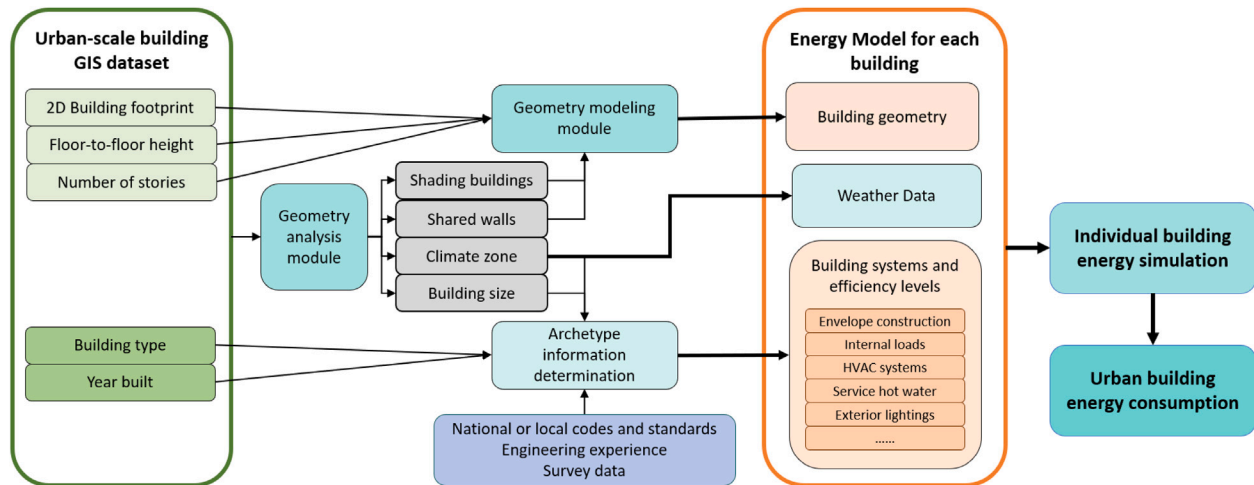


Fig. 4. Major data sources and workflow of the urban building energy modeling.

predictions with observed data and on any existing prior knowledge. In particular, Bayesian calibration has been increasingly used for BEM calibration because of the following advantages over deterministic approaches [6,130,193,196]:

- Its ability to naturally incorporate expert knowledge derived from a collection of data sources (e.g., drawings, technical specifications, surveys, literature, and industry standards) into the calibration process.
- It is particularly beneficial when measured data are sparse, as the inclusion of prior information can help guide the estimation of uncertain parameters even when observations alone might be insufficient to fully constrain the model [130].
- BEM calibration is an ill-posed inverse problem that often suffers from parameter nonidentifiability because different combinations of parameter sets can yield similar goodness of fit with measured data. Providing a distribution of possible parameter values rather than a single best-fit value therefore instills greater confidence in the model's predictions.
- Through Bayesian inference, the model can be updated as new data become available, providing a way to iteratively approach to continuously improve the model's credibility.

The Bayesian inference process for calibration involves: (1) specifying prior distributions for the uncertain parameters; (2) defining a likelihood function that computes the probability of observing the measured data given a set of parameter values; and (3) applying Bayes' theorem to combine the prior and likelihood to obtain updated (posterior) distributions for the uncertain parameters after observing data. The choice of priors is critical in Bayesian inference, as poorly chosen priors can bias calibration results [197]. Therefore, using weakly informative priors has been recommended over highly constrained ones to avoid excluding plausible parameter values while still achieving some useful regularization to guide the calibration [130].

Since posterior distributions usually cannot be derived analytically for complex building energy models, Markov Chain Monte Carlo (MCMC) methods are often used to sample from these distributions [130]. However, Bayesian inference via MCMC can be computationally intractable when the likelihood function involves BEM simulations due to the large number of simulations required to achieve convergence to a stationary distribution. Approaches to reduce computation costs include using meta-models [131,132], utilizing more efficient MCMC sampling strategies [129,198], and employing Approximate Bayesian

Computation to avoid the need for explicit likelihood evaluations [135, 199].

2.8. Question 8: How does model calibration differ between individual buildings and urban scale?

Urban building energy modeling (UBEM) refers to creating physics-based building energy models for each physical building in an urban area [200]. UBEM involves varying spatial and temporal resolutions. For the spatial scale, UBEM can cover dozens of buildings within a community, thousands of buildings within a district, or tens of thousands of buildings across an entire city. For the temporal scale, hourly or sub-hourly simulations are typically performed. Unlike a single building model, it is difficult to obtain detailed information for the building stock of a city; therefore, UBEM typically relies on publicly available GIS-based building basic information datasets as a major input source. As shown in Fig. 4, the building footprint polygon, floor-to-floor height, and number of stories are typically used to create the geometry for the target building, and their surrounding buildings are modeled as shading buildings [201]. The building systems and energy efficiency levels are typically inferred from archetype information, which was determined based on climate zone, building type, building size, and/or year built [202]. The buildings that belong to the same archetypes have the same settings for envelope construction, windows-to-wall ratio, internal loads, HVAC systems, and service hot water systems [203]. The archetype information is typically created based on literature review, national or local codes and standards, and engineering experience [204].

For UBEM calibration, the system efficiency levels and operational settings are typically tuned to minimize the gaps between the simulated and measured energy use. The parameters to adjust usually include the envelope construction (walls, roofs, windows) properties, internal heat gain intensities, HVAC system efficiencies, window-to-wall ratio (WWR), air infiltration, HVAC operation schedules, setpoints of indoor air cooling and heating temperatures, and occupancy schedules.

At the urban scale, measured energy use or other monitored building performance data are limited. The typical available data include annual energy use (electricity and natural gas) for a group of commercial buildings subject to a city's building energy benchmarking or disclosure regulation requirements. More granular data, such as monthly energy use (e.g., from EnergyStar Portfolio Manager) or hourly energy use (from smart meters), is even more difficult to obtain (sometimes due to privacy concerns). Sometimes, aggregated annual energy

Table 3
Summary of calibration methods used in urban building energy modeling (UBEM).

Calibration method	Parameters	Measured data	Time scale	Performance metrics	No. of buildings	Ref.
Building-by-building	CST, WWR, WUV, WST, INF	Elec., WiFi-OCC, Weather	Hourly	NMBE 10%, CVRMSE 30%	35–100	[205]
Building-by-building	13–20 common parameters	Steam, Elec., Chilled Water	Monthly/ Annual	PE	100	[206,207]
Learning from prototypes	17 common parameters	EUI, GUI, Heating Intensity	Annual	GOF	72–483	[180,208]
Archetype info	LPD, STP, OCC, MLT	EUI	Annual	Building level: PE District level: Kolmogorov–Smirnov statistic	164	[209]

Note. CST = construction type; WWR = window-to-wall ratio; WUV = window U-value; WST = wall insulation; INF = infiltration; EUI = energy use intensity; GUI = gas use intensity; GOF = goodness of fit; PE = prediction error; LPD = lighting power density; STP = setpoint; OCC = occupancy; MLT = miscellaneous loads.

use data of buildings are available at certain spatial levels, such as city block, census tract, or zipcode (e.g., the Energy Atlas for California, <https://www.energyatlas.ucla.edu/>).

The UBEM calibration methods can be organized into three categories as shown in Table 3.

The first is the building-by-building approach, which uses the same calibration method and directly scales the calibration from one building to many buildings [205]. This approach requires data available for each building and requires a significant amount of computational resources.

The second approach is to learn from prototype building energy models or previously simulated results to speed up the calibration process. As introduced above and illustrated in Fig. 4, the energy models in the UBEM have different geometries; however, the system types and energy efficiency values are typically the same for a given archetype. For example, all large office buildings in the UBEM could have chillers, natural gas boilers, and central variable air volume systems for space cooling and heating [180]. Those buildings may perform similarly when the system efficiency values change from one to the other [210]. For instance, when we reduce the lighting power density and increase the coefficient of performance (COP) of the chillers, the simulated energy consumption of all those buildings decreases. In this approach, prototype building with different sets of efficiency values could be performed to create a common dataset with detailed simulation results for each set of efficiency values. The majority of the buildings can learn from this prototype building dataset to reduce the number of calibration runs based on this similarity.

The third method is archetype calibration, which calibrates the archetype parameters using the measured data from a group of buildings and applies the calibrated archetype parameters to all buildings [209]. This method uses probabilistic/Bayesian calibration to obtain the posterior distribution of calibrated parameters as shown in Fig. 5.

For the calibration of individual buildings in the UBEM, as those building energy models are created using the same procedure and tools, there are some connections among the simulated results. The calibration process at the urban scale should leverage structural similarities among buildings to enhance speed and efficiency, streamlining the process and reducing computational requirements. For the calibration of city-scale UBEM, it may not be necessary to calibrate each individual building separately. Instead, we should compare the aggregated performance metrics and their distributions. Depending on the use case, different performance metrics should be used to guide the UBEM calibration. For example, when a city tries to evaluate and prioritize energy efficiency retrofits for its building stock, the annual total energy use and energy costs can be the metrics, while for a utility's distribution system planning, the coincident peak electricity demand at the feeder or substation level should be the metrics.

There are challenges for UBEM calibration. First, the available measured energy use data for urban buildings are very limited. Few buildings have accurate and higher-resolution measured performance data. The second challenge is that the computational resources needed

for urban-scale model calibration are substantially higher than for individual building calibration, especially with the need to consider the urban context of individual buildings. Consequently, individual building calibrations can employ computationally intensive methods, such as Bayesian and deep learning. However, directly applying these methods across all buildings at the urban scale is computationally impractical. The third one is the weather data used for UBEM calibration; most of the studies use TMY weather data. For those using measured weather data, usually only data from one weather station was used to represent the ambient conditions of a large urban area. As the price of the Internet of Things (IoT) has reduced, more and more localized weather data is publicly available, which could be used to calibrate the UBEM.

2.9. Question 9: How can recent advancements in AI/ML enhance model calibration?

Recent advancements in AI/ML are reshaping model calibration by integrating data-driven techniques with conventional physics-based models. These advances address key challenges in calibration, such as computational intractability, model discrepancies, data scarcity, and the high-dimensional nature of parameter tuning. Broadly, AI/ML enhancements in calibration can be categorized into four areas:

- **Surrogate or meta-modeling:** AI/ML models are trained to emulate the behavior of BEM, offering computationally efficient alternatives to full-scale energy simulations.
- **Discrepancy Modeling:** AI/ML models learn the residuals (i.e., the differences between simulation outputs and actual measurements) to enhance model accuracy.
- **Data Augmentation:** AI/ML methods generate synthetic data to augment existing datasets, mitigating challenges related to data scarcity and quality.
- **Data Acquisition:** AI/ML techniques optimize data collection by identifying the most informative data points, improving efficiency and minimizing redundancy.

While still not universally adopted, surrogates are not new to calibration workflows, and their ability to accurately emulate simulation models has been demonstrated, particularly in Bayesian and optimization-based methods [117,131]. Hybrid approaches that use data-driven methods to predict residual errors in physics-based models have also been investigated, showing improvements in prediction accuracy [211–213]. In some cases, a discrepancy term may be incorporated into the surrogate to account for deviations between simulation models and actual physical systems [126,214].

Despite their benefits, data-driven surrogate and residual modeling approaches often lack interpretability and generalization. These models tend to be case-specific, performing well on the building and conditions they were trained on but struggling to generalize into unexplored regimes [215]. Recent advancements such as explainable AI (XAI), physics-informed neural networks (PINNs), graphical neural networks

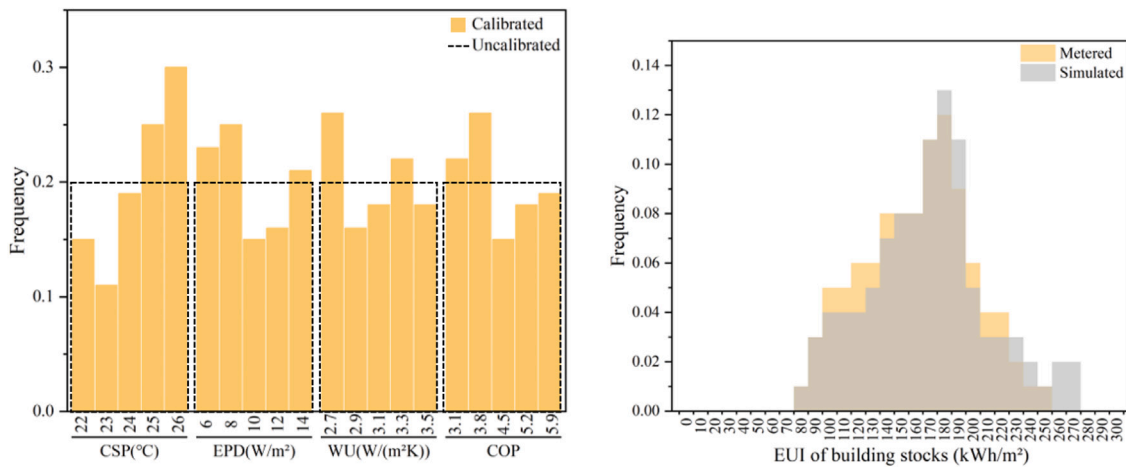


Fig. 5. Calibration results of different UBEM methods.

(GNN), and neural operators offer potential pathways to addressing these challenges.

To improve the interpretability and transparency of complex AI/ML models, post-hoc explainability methods such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) can be used to generate insights into how these models make predictions [216,217]. PINNs, or more generally physics-informed machine learning, can enhance generalizability in solving ill-posed inverse problems by embedding physical laws into machine learning models, ensuring physically consistent predictions even for extrapolatory tasks [218]. Similarly, Graph Neural Networks (GNNs), when integrated with physics-based inductive biases, have demonstrated enhanced predictive performance [219]. By representing system components as nodes and their interactions as edges, graphs can provide a flexible and scalable framework for modeling dynamic systems while allowing physics-based inductive biases to be embedded into pairwise interactions [220]. Finally, neural operators also offer a promising avenue for generalizable inverse modeling by learning mappings between function spaces, rather than relying on discretized input–output data for training [221]. While these models are primarily developed to enhance predictive accuracy and generalizability by incorporating physical structure, they also offer a promising strategy for inverse modeling. Their differentiable architecture, combined with the ability to learn the underlying dynamics of physical systems by design, enables efficient parameter estimation in high-dimensional, complex systems where traditional calibration methods often face limitations due to computational intractability and uncertainty [222].

While recent advancements in BEM calibration have focused on improving calibration methods [6,13], it is important to recognize that BEM calibration remains fundamentally data-centric. As an ill-posed inverse problem, calibration is prone to over-parameterization, where multiple parameter combinations can yield similar goodness-of-fit, making the identification of a unique solution challenging. Although controlled experiments can generate high-quality data for accurate calibration [50], such datasets are often impractical to obtain in operational buildings. While building management systems (BMS) enable large-scale data collection, this data may be less informative than its volume suggests, as buildings typically operate under fixed setpoints [129].

To address these challenges, methods such as generative adversarial networks (GANs) can augment operational data when available data is scarce or lacks variability while also introducing an extra layer of data protection by generating synthetic datasets without direct linkage to the original data sources [223,224]. While GANs focus on data augmentation, active learning improves data efficiency by selecting

the most informative data points for calibration, reducing the need for extensive real-world data collection and leveraging diverse, high-entropy data to better identify uncertain calibration parameters. By focusing data collection on regions of high uncertainty, this targeted approach has been shown to produce more generalizable models while minimizing redundant data [225,226]. However, a challenge with active learning is that certain “informative” experiments may disrupt occupants or interfere with normal building operations. One way to mitigate this is by incorporating a cost metric that penalizes deviations from normal operations, allowing the algorithm to balance information gain with real-world constraints [226]. Additionally, Active Learning can also be strategically deployed during unoccupied periods, such as commissioning or maintenance, to conduct experiments without disrupting operations.

Despite recent advancements, calibrating BEMs remains labor-intensive and often requires substantial domain expertise. LLMs, along with other generative models such as GANs, a form of generative AI, offer new opportunities to streamline BEM calibration [227]. These include interpreting simulation and measured data, recommending parameter adjustments, and generating building models. A promising direction is a bilevel approach [228], in which an LLM generates hypotheses about model structure or parameters, which are then translated into BEMs, simulated, and refined based on calibration performance feedback. Recent work demonstrated that LLMs can translate natural language building descriptions into template-like simulation models [229]. However, accurate calibration at the building scale requires moving beyond templates toward generating realistic geometry and system configurations. This will likely require multimodal LLMs [230] capable of integrating diverse inputs (such as as-built drawings, component specifications, and operational data) into simulation-ready representations. Moreover, the environmental cost of training LLMs should not be ignored [231], which is somewhat contradictory given that BEM calibration underpins efforts to reduce building energy consumption.

The AI/ML methods discussed here are not exhaustive, and other emerging techniques may further enhance BEM calibration. However, integrating these approaches into practical workflows requires further research on scalability, reliability, and compatibility with existing BEM frameworks. Future efforts should focus on bridging the gap between theoretical advancements and real-world applications to ensure these methods can be effectively deployed in operational buildings. Human oversight remains essential, as overreliance on AI risks overlooking critical value judgments and may lead to factual inaccuracies [232].

2.10. Question 10: What are the current challenges and future perspectives in model calibration?

Current challenges:

- There is currently no consensus on when a model can be considered “calibrated”. While CVMSE and NMSE have been widely adopted, it is obscure which variables or scale they should be evaluated on. Many statistical methods can obtain a model that meets the criteria and is considered calibrated based on the standards. The broader usefulness of such models is seldom demonstrated, and whether agreement on one or two outputs is sufficient for practical applications remains unanswered [196].
- Current guidelines with CVMSE and NMSE thresholds allow multiple sets of calibration parameter values to meet the criteria due to overparameterization, leading to identifiability issues [109]. These issues could be mitigated by constraining the parameters, calibrating against more outputs, and using hybrid methods such as multi-stage or ensemble calibration. However, such processes are highly expert-driven due to the complexity and heterogeneity of buildings.
- Existing calibration studies often assumed that the necessary information and measured data are available with reliable quality. However, it is practically difficult and time-consuming to fulfill the data requirements [233]. Acquiring and processing the operational data is typically a long process. While dedicated experiments are necessary for parameters such as infiltration rate, others, such as envelope thermal properties, are almost impossible to measure accurately.
- BEM models inevitably contain some sort of assumptions, abstractions, and approximations. This model discrepancy does not just stop the modelers from obtaining a “perfect” model but also affects the calibration of uncertain parameters [234]. While Bayesian-based methods explicitly account for this uncertainty, it is not straightforward to distinguish the model form uncertainty from other sources of uncertainty.
- Existing work on BEM calibration is often difficult, if not impossible, to reproduce, with evaluations of proposed calibration algorithms frequently based on unique building case studies [6]. This lack of transparency and absence of benchmarks makes it challenging to build upon previous work and assess the trustworthiness and generalizability of proposed methods.

Future perspectives:

- The increasing interest in digital twins highlights the advantage of building energy simulation models in being able to generalize beyond the training data. In this context, calibration plays an important role in continuously assimilating measured data and keeping the model a real-time counterpart of the actual buildings [136].
- The other side of the bidirectional information exchange of digital twins is to contribute to the operation and management. The calibration and evaluation should be conducted regarding certain applications. With the absence of a universal metric, it is critical to evaluate the fit-for-purpose performance of calibrated models.
- Advancements in AI/ML methods offer promising avenues to enhance BEM calibration by improving data efficiency, automating workflows, and supporting model development. While techniques such as GANs, active learning, and LLMs show potential, they are not exhaustive, and further research is needed to ensure scalability, reliability, and integration with existing practices. Human oversight, however, remains essential to ensure that these technologies are applied appropriately and with due consideration for context-specific judgment.

- To reproduce and benchmark future calibration studies and promote consistency in standards, there is a strong need to collect or develop open-sourced calibration datasets and models. In addition to some initial efforts [235,236], each building case should include static information and measured data with high granularity, original base models, and calibrated models that are verified to match the ground truth as much as possible. Eventually, the model set should cover diverse building types, HVAC systems, efficiency levels, as well as climate zones.

3. Conclusion

Through this ten questions paper, we have examined the motivations, evaluation metrics, data requirements, methods, and emerging directions of BEM calibration. A central message is that calibration is inherently contextual and purpose-driven. Rather than adhering to fixed thresholds or generic workflows, effective calibration requires modelers to critically navigate trade-offs between model complexity, data availability, computational resources, and stakeholder needs. The “right” calibration approach depends on the specific question being addressed and the decisions it is intended to inform.

Calibration should not be viewed as merely a technical process aimed at satisfying prescribed thresholds, such as those in ASHRAE Guideline 14, as this risks instilling false confidence that the model is “good enough”, while potentially obscuring underlying limitations that may result in misguided conclusions. Instead, the evaluation of calibrated models should be context-dependent, with the selection of simulation outputs, their temporal and spatial resolution, and the corresponding performance metrics all aligned with the model’s intended application. While recent advances in AI and machine learning offer promising opportunities to automate and augment the calibration process, further work is needed to integrate these approaches into practical workflows. Bridging the gap between theoretical methods and real-world deployment remains a key challenge.

Finally, the paper highlights the urgent need to improve reproducibility and comparability across calibration studies. Open-access datasets, reference models, and standardized benchmarking protocols are essential to move the field beyond isolated case studies toward more generalizable and credible practices. Transparent documentation is equally critical in supporting this shift. As building simulation plays an increasingly central role in retrofit analysis, smart operations, climate mitigation and adaptation, and digital twins, the development of robust, transparent, and context-aware calibration practices will remain fundamental to realizing the full potential of BEM.

CRedit authorship contribution statement

Adrian Chong: Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Da Yan:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Kaiyu Sun:** Writing – review & editing, Writing – original draft. **Sicheng Zhan:** Writing – review & editing, Writing – original draft. **Siyu Cheng:** Writing – review & editing, Writing – original draft, Visualization. **Yi Wu:** Writing – review & editing, Writing – original draft. **Yixing Chen:** Writing – review & editing, Writing – original draft. **Tianzhen Hong:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A.1
Three main categories of building energy model (BEM) calibration methods.

Category	Evidence-based	Optimization-based	Calibration under uncertainty
Typical calibration methods	Energy audit, interview, in-situ measurement, etc.	Particle swarm optimization, genetic algorithm, Bayesian optimization, etc.	Markov chain Monte Carlo, Hamiltonian Monte Carlo, etc.
Modeling approaches	BEM (e.g., EnergyPlus, TRNSYS, DeST)	BEM or surrogate models (e.g., Gaussian Process, linear regression)	BEM and surrogate models
Uncertainty awareness	No	No	Yes
Data requirements: static	As-built drawings, system specifications, etc.	Not required, but may be used to define the base model.	
Data requirements: dynamic inputs	Boundary conditions used for output comparison (e.g., actual weather); often transformed into parameters such as nominal loads, schedules, and setpoints (e.g., electricity use, zone temperatures, setpoints, occupancy, etc.)		
Data requirements: dynamic outputs	Optional for calibration, often used for model evaluation	Electricity use, heating/cooling loads, zone temperatures, etc.	
Temporal resolution	Typically based on short data snippets or low-resolution (daily/monthly) output comparisons	Requires high temporal resolution (hourly or sub-hourly) to improve parameter identifiability	Adaptable to various resolutions, but excessive data length may result in intractable problems
Spatial resolution	Parameter values may correspond to either zone-level details or abstract overall properties depending on available data. Optimization and uncertainty-based methods can incorporate multiple outputs (e.g., sub-metered data) into objective functions, although the latter significantly increases computational cost.		
Calibrated parameters	Optimization and uncertainty-based approaches generally cannot handle large numbers of parameters in a single calibration task. Sensitivity analysis may be applied to support parameter screening and improve identifiability.		
Expert knowledge dependency	Manual parameter specification based on expert judgment	Automation requires expert input to define parameter bounds, data sources, and objective functions	Expert knowledge is needed to define prior distributions, construct datasets, monitor calibration, and interpret posterior results
Computational cost	Low; may not require iterative simulation	Medium; depends on optimization algorithm and parameter count	High; influenced by sampling method, sample size, and number of parameters
Key advantages	Parameters derived from physical insights; more representative	Flexible and relatively easy to configure; less dependent on expert interpretation	Integrates data and expert knowledge while explicitly accounting for uncertainty
Key limitations	Limited by data availability; high accuracy is difficult to achieve	Susceptible to identifiability issues; parameter values may be unreliable	Computationally demanding; requires high-quality data and substantial expert involvement

Table A.2
Open-source datasets for BEM calibration.

Data type	Sub-type	Name	Description	Time resolution	Ref.
Weather	Actual meteorological year (AMY)	ERA5	Atmospheric reanalysis dataset from 1970 to present	Hourly	[158]
		MERRA-2	Atmospheric reanalysis dataset from 1979 to present	Hourly	[159]
	Short-term weather API	MeteoBlue	APIs for weather forecasts and historical data	Hourly	[237]
		Open Weather	APIs for current weather, forecasts, and historical data	Sub-hourly (1 min)	[238]
		Open Meteo	APIs for hourly forecasts at 1 km mesoscale	Hourly	[239]
Prototype building	United States	DOE prototype building	16 commercial types and 2 residential prototypes in the US	–	[160]
	China	Chinese prototype building	4 residential and 11 commercial buildings, with 9225 models in China	–	[161]
	Europe	Tabula	Residential building typologies from 20 European countries	–	[162]
Internal gains	Occupant behavior (OB) dataset	ASHRAE Global OB database	Occupancy patterns and behavior data collected worldwide	Sub-hourly	[163]
		Center for Time Use Research	Collections of time-use diary data in multiple countries	Not fixed	[240–242]
		ROBOD	181 days of occupancy data from five rooms at 5-minute resolution	Sub-hourly (5 min)	[165]

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Table A.2 (continued).

Data type	Sub-type	Name	Description	Time resolution	Ref.
Benchmarking	Energy-use data	ASHRAE – Great Energy Predictor III	Metered building energy usage data from over 1,000 buildings over a 3-year period	Hourly	[49]
		Building Performance Database (BPD)	Energy-related characteristics and yearly usage data of buildings in the US	Yearly	[243]
	Cross-sector data	Benchmarking dataset for buildings	Building operational and indoor/outdoor environmental data	Sub-hourly	[167]
		Pecan Street dataset	Residential energy use data across building, transportation, and agriculture sectors	Not fixed	[168]
		Open Energy Data Initiative (OEDI)	Geothermal energy data, plus power generation and transportation data	Not fixed	[169]

Note. “–” indicates the time resolution is not explicitly specified.

Table A.3

Examples of evaluation metrics for building energy model calibration.

No	Metric	Formula	Remarks
<i>Point Distance Accuracy Measures</i>			
1	Maximum Absolute Difference (MAD)	$\max(M_i - S_i)$	Largest absolute deviation between simulated and measured values. Scale-dependent and sensitive to outliers.
2	Peak Error (PE)	$ M_{\max} - S_{\max} $	Maximum deviation between predicted and observed peak values. Scale-dependent and highly sensitive to outliers.
3	Normalized Peak Error (NPE)	$\frac{PE}{M_{\max}} \times 100\%$	Peak Error normalized by the maximum measured value. Expressed as a percentage and dimensionless.
4	Root Mean Squared Error (RMSE)	$\sqrt{\frac{1}{n} \sum_{i=1}^n (M_i - S_i)^2}$	Scale-dependent metric with the same unit as the prediction. Sensitive to outliers.
5	Root Mean Squared Logarithmic Error (RMSLE)	$\sqrt{\frac{1}{n} \sum_{i=1}^n (\log(M_i + 1) - \log(S_i + 1))^2}$	RMSLE is a logarithmic variant of RMSE that considers relative errors and has reduced sensitivity to large absolute errors.
6	Geometric RMSE (GRMSE)	$\sqrt[n]{\prod_{i=1}^n (M_i - S_i)^2}$	Geometric variant of RMSE with reduced sensitivity to outliers. Harder to interpret and remains scale-dependent.
7	Coefficient of Variation of RMSE (CVRMSE)	$\frac{RMSE}{M} \times 100\%$	Dimensionless form of RMSE, used in ASHRAE Guideline 14. Sensitive to small means.
8	Normalized Mean Bias Error (NMBE)	$\frac{\sum_{i=1}^n (M_i - S_i)}{nM} \times 100\%$	Dimensionless and scale-dependent. Indicates error direction; used in ASHRAE Guideline 14.
<i>Time-Series Similarity Metrics</i>			
9	Normalized Cross Correlation (NCC)	$\frac{\sum_{i=1}^{n-k} (M_i - \bar{M})(S_{i+k} - \bar{S})}{\sqrt{\sum_{i=1}^{n-k} (M_i - \bar{M})^2} \sqrt{\sum_{i=1}^{n-k} (S_{i+k} - \bar{S})^2}}$	Measures similarity between two time series with a lag k . Interpretable but sensitive to time shifts.
10	Fourier Transform Distance (FTD)	$\sqrt{\sum_{i=0}^K (F(M)_i - F(S)_i)^2}$	Compares frequency-domain representations. Less sensitive to time shifts but may overlook local variations.
11	Dynamic Time Warping (DTW)	$\min_P \sum_{(i,j) \in P} \sqrt{(M_i - S_j)^2}$	Aligns time series with nonlinear warping. Effective for distortions, but computationally expensive and sensitive to noise.
12	Lower Bound Keogh DTW [244]	$\sum_{i=1}^n \begin{cases} (S_i - U_i)^2, & S_i > U_i \\ (L_i - S_i)^2, & S_i < L_i \\ 0, & \text{otherwise} \end{cases}$	An efficient approximation of DTW that constrains warping using an envelope. Less sensitive to noise and faster to compute.

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Table A.3 (continued).

No	Metric	Formula	Remarks
<i>Probability-Based Evaluation Metrics</i>			
13	Kullback–Leibler Divergence	$\int_{-\infty}^{\infty} M(x) \log \frac{M(x)}{S(x)} dx$	Measures dissimilarity between probability distributions. Asymmetric and unbounded. See [61] for alternatives such as Jensen-Shannon divergence.
14	Continuous Ranked Probability Score (CRPS)	$\int_{-\infty}^{\infty} (F_S(x) - F_M(x))^2 dx$	Compares simulated and measured CDFs. Useful for evaluating probabilistic forecasts.
15	Single Observation CRPS	$\int_{-\infty}^{\infty} (F_S(x) - \mathbf{1}_{x \geq m})^2 dx$	Compares a predicted CDF against a single observation. Useful for event-wise validation.
16	Prediction Interval Coverage Probability (PICP)	$\frac{1}{N} \sum_{i=1}^N \mathbf{1}(L_i \leq m_i \leq U_i)$	Measures the proportion of observed values within predicted intervals. Often used with the Coverage Width Criterion (CWC) to balance reliability and sharpness [245].

Table A.4

Summary of commonly used measurement methods for observable model inputs.

Measure- ment	Thermal resistance and transmittance (BS ISO 9869-1)	Air infiltration and air change rate (BS EN ISO 12569)	Air permeability of building envelope (BS EN ISO 9972:2015)	Occupant sensing and data collection (IEA EBC Annex 66)
Method description	Heat flow meter method: 1. Average method (Sec. 7.1) 2. Corrected average method (Sec. 7.2) 3. Dynamic method (Annex B)	Tracer gas dilution method: Concentration decay (Sec. 4.2) Continuous dose (Sec. 4.3) Constant concentration (Sec. 4.4)	Fan pressurization method	Experimental methods (Sec. 4.1): In-situ studies, laboratory studies, surveys, and mixed methods
Instruments / tools	Heat flux meter, surface and air temperature sensors, data loggers	Tracer gas cylinder with flow meter, dosing device, gas analyzer, sampling tubes	Air-moving fan, pressure sensors, airflow meters, barometer, sealing materials	Occupant sensing and data acquisition technologies (Section 4.2) Direct sensing tools: Infrared motion sensors (PIR), ultrasonic sensors, video cameras, wearable sensors. Indirect sensing tools: CO ₂ concentration sensors, power meters / plug load sensors, Wi-Fi/Bluetooth device tracking.
Measure- ment procedures	Mount heat-flow meter with thermal paste. Install temperature sensors on both sides. Ensure a sufficient ΔT . Continuously measure and record data at fixed intervals. The test duration must satisfy the following criteria: Duration: At least 72 hours of data. Convergence: Thermal resistance change over the past 24 hours is less than 5%. Consistency: Values from the first and last two-thirds of the data differ by less than 5%.	Prepare the test environment (seal space if needed). Inject and mix tracer gas uniformly. Monitor decay over time until the concentration has decayed sufficiently. Data analysis: Plot the decay curve (concentration vs. time); Calculate air change rate.	Prepare the building. Install blower door. Use the blower door fan to pressurize and depressurize the building at multiple pressure levels. Measure the airflow rate required to maintain that pressure. Data analysis: use the power law regression to relate airflow to pressure difference.	Determine the occupant behavior to be studied and influencing factors. Specify spatial resolution (zone / room / seat level) and temporal resolution (real-time, sub-hourly, hourly, daily). Select measurement methods and sample size. Obtain ethics clearance. Determine sensors and data-logging infrastructure. Test and install sensors. Conduct pilot study. Data logging and storage. Data processing and analysis.
Data analysis	R-value: $R = \frac{\sum (T_{si} - T_{se})}{\sum q_i}$ C-value: $C = \frac{\sum q_i}{\sum (T_{si} - T_{se})}$ Use dynamic method if high thermal mass exists.	Two-point method: $n = \frac{1}{t_1 - t_0} \cdot \ln \left(\frac{C(t_0)}{C(t_1)} \right)$ Multi-point method: $n = -\text{slope of } \ln(C(t)) \text{ vs. } t$	Power-law model: $Q = C \cdot (\Delta P)^n$ Log fit: $\log(Q) = \log(C) + n \log(\Delta P)$ Compute air permeability (m ³ /h·m ²) and air change rate (h ⁻¹) at reference pressure: $q_{50} = \frac{Q_{50}}{A_{env}}, \quad n_{50} = \frac{Q_{50}}{V}$	Depends on method. Often used for validation, clustering, or deriving schedules from time-series data.
Notes	Avoid thermal bridges and direct solar radiation (or other heat sources). Use night-time data for glazing. Encourage strong ΔT . Validate the results by checking for convergence and consistency.	Choose suitable tracer gas. Ensure well-mixed gas distribution. Minimize external influence.	Seal openings and turn off HVAC. At least 5 valid pressure steps (10–100 Pa) are required for curve fitting. Report if unable to reach 25 Pa. Let the pressure stabilize before recording each flow rate. Record temperature, pressure, and wind conditions. Validate with multiple measurements if needed.	Methods vary in accuracy, spatial, and temporal resolution. Be aware of the trade-offs associated with different methods. Consider privacy and ethics. Combine sensing with diaries or surveys for context.

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Table A.4 (continued).

Measure-ment	Thermal resistance and transmittance (BS ISO 9869-1)	Air infiltration and air change rate (BS EN ISO 12569)	Air permeability of building envelope (BS EN ISO 9972:2015)	Occupant sensing and data collection (IEA EBC Annex 66)
Advantages	Simple tools, real-world accuracy, usable for energy audits, retrofit, and verifying insulation performance	Reflect real building behavior, low-cost, suitable for natural ventilation studies	Standardized, repeatable, and fast; useful for diagnostics, benchmarking, and comparing airtightness across buildings	In-situ studies Realistic behavior, Long-term deployment, Supports model validation Laboratory studies Controlled environment, support causal understanding Surveys Scalable, cost-effective, captures subjective perception
Limitations	Requires long measurement duration, steady-state required, sensitive to placement and weather	Requires uniform mixing, limited dynamic insight, weather-dependent	Not representative of real-world infiltration, requires building to be unoccupied and sealed, expert setup needed, limited in large buildings	In-situ studies Limited control, complex setup, privacy concerns Laboratory studies May not reflect real use, small sample size, high cost Surveys Recall and response bias, low temporal precision, dependent on design

Data availability

Data will be made available on request.

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